

CSE460

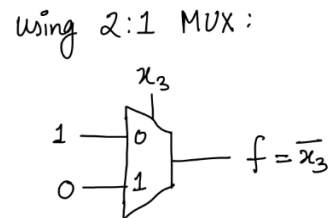
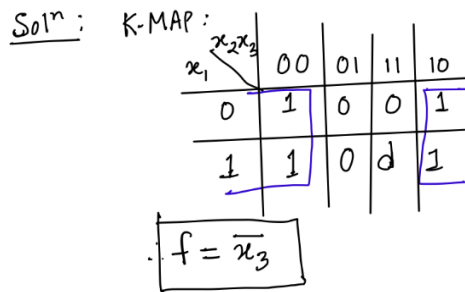
Practice Sheet

CMOS Logic

Q-1. Suppose you want to design a circuit that includes three inputs (x_1, x_2, x_3), and one output (f). Following is the truth table of the desired circuit.

x_1	x_2	x_3	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	d

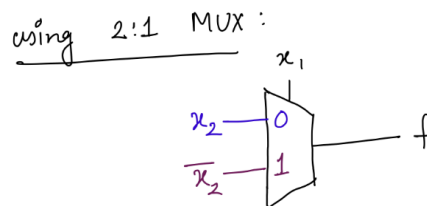
Find the logic expression using k-map and implement the logic expression using only 2x1 mux (Don't use any logic gate; you can use more than one 2x1 mux).



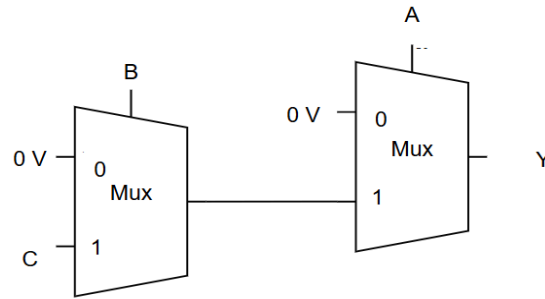
Q-2. Implement an XNOR gate using a single 2x1 mux.

Solⁿ: XNOR Gate $\rightarrow f = x_1 \oplus x_2 = \overline{x_1}x_2 + x_1\overline{x_2}$

x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	0



Q-3. Form a truth table and identify the logic expression of the given circuit.



Solⁿ:

Truth Table:

A	B	C	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

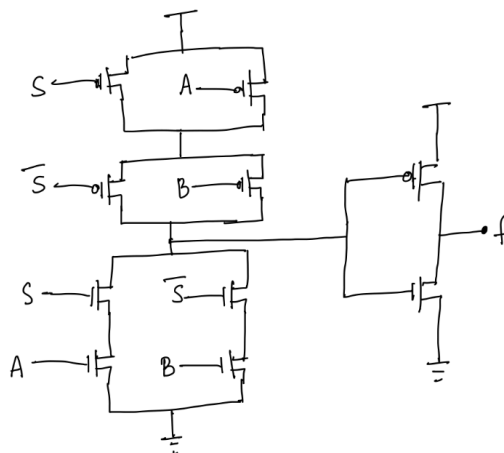
A \ BC	00	01	11	10
0	0	0	0	0
1	0	0	1	0

$\therefore f = ABC$

Q-4: Compare the number of required transistors for implementing a 2x1 mux using CMOS and transmission gates. (Show both implementation diagram)

Solⁿ:

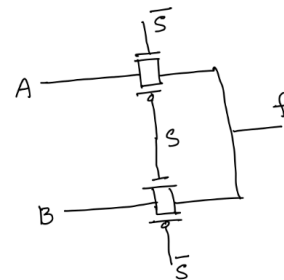
CMOS:



No. of transistors = 10

$$f = SA + \bar{S}B$$

Using Transmission Gate:



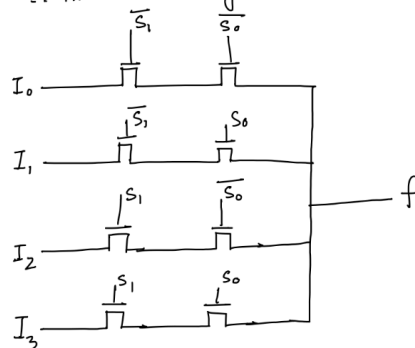
No. of transistors = 4

Q-5. Design a 4x1 Mux using minimum number of nmos pass transistors. You are only given the input signals s, D0, D1, D2, D3. Using the 4x1 Mux, design a 8x1 Mux(you cannot use any other Mux).

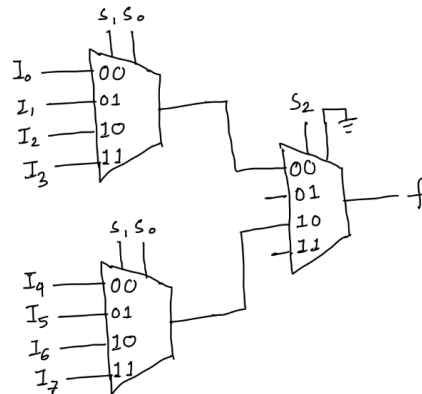
Solⁿ:

s_1	s_0	f
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

∴ 4x1 Mux using Pass Transistors:



8x1 Mux using 4x1 Mux:



Q-6. Take a room light control system for example. The light should turn ON automatically only:

- if the main power switch is ON
- and either the wall switch is ON or a person is detected by the motion sensor

Take the inputs as following:

- **A** = Main power switch (1 = ON, 0 = OFF)
- **B** = Wall switch (1 = ON, 0 = OFF)
- **C** = Motion sensor (1 = detects movement, 0 = no movement).

(a) Find the expression for output **f** (1 = Light ON, 0 = Light OFF).

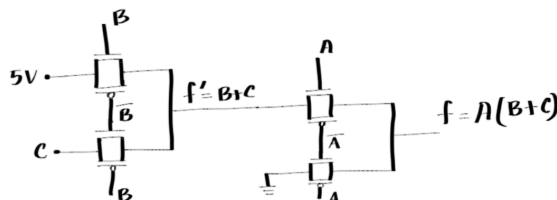
(b) Design the circuit using transmission gates.

$$a) f = A \cdot (B + C)$$

$$b) f = A \cdot (B + C) = A \cdot f'$$

B	C	f'
0	0	0
0	1	1
1	0	1
1	1	1

A	f'	f
0	0	0
0	1	0
1	0	0
1	1	1



Q-7. For the following truth table:

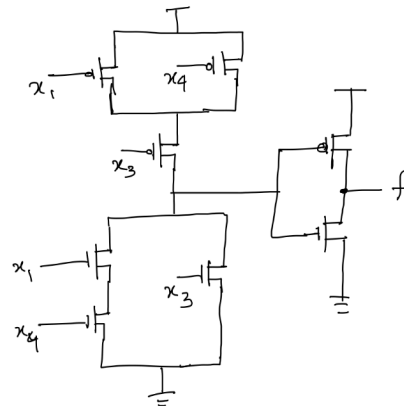
X1	X2	X3	X4	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	d
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	d

- A) Find the expression for f using K-map.
 B) Now, design the circuit using CMOS logic.

Solⁿ:

x_3x_4	00	01	11	10	
x_1x_2	00	0	0	1	1
01	0	0	1	1	
11	0	1	d	1	
10	0	1	d	1	

$$f = x_3 + x_1x_4$$



Q-8. Write truth table of a 4 to 1 mux whose select pins are A, B and Input lines are I_0, I_1, I_2, I_3

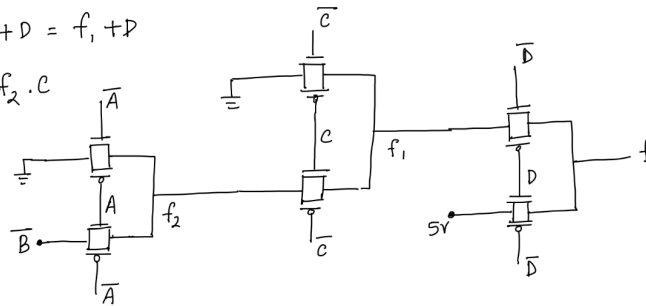
Q-9. Implement the following function using only transmission gate logic:

$$f = A \cdot \overline{B} \cdot C + D$$

Solⁿ: $f = A \cdot \overline{B} \cdot C + D = f_1 + D$

$$f_1 = A \cdot \overline{B} \cdot C = f_2 \cdot C$$

$$f_2 = A \cdot \overline{B}$$



Q-10. Find the simplified output logic expression $Y = F(A, B, C)$ using **karnaugh's map** rule

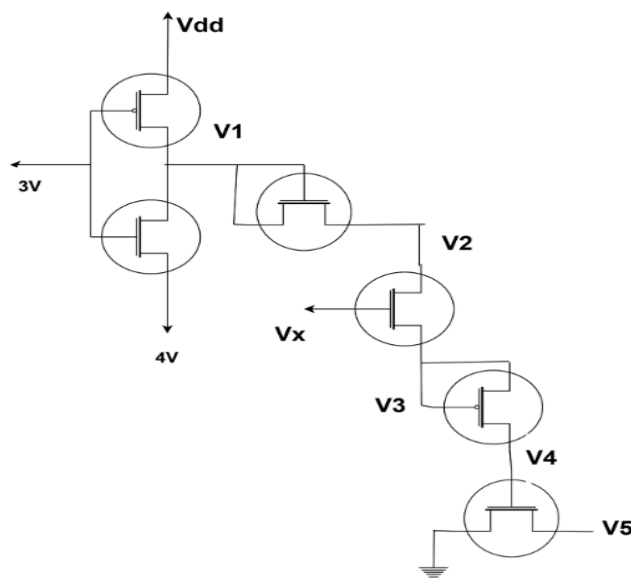
AB \ C	00	01	11	10
0	1	0	0	1
1	1	1	0	1

Solⁿ:

AB \ C	00	01	11	10
0	1	0	0	1
1	1	1	0	1

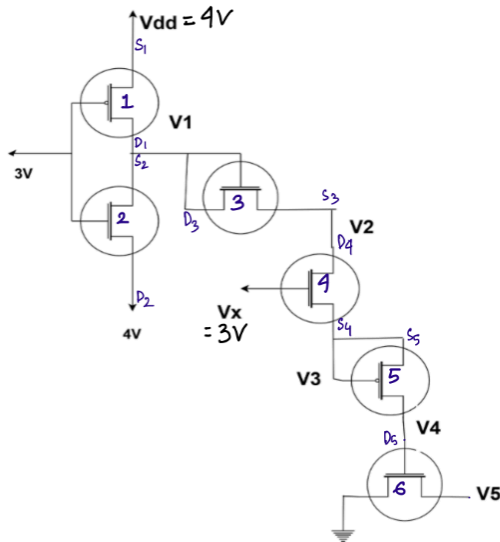
$\therefore Y = \overline{B} + \overline{A}C$

Q-11. Find out the voltages at V1, V2, V3, V4, V5 when $V_{DD} = 4V$, and $V_x = 3V$, $V_{tn} = 0.5V$, $|V_{tp}| = 0.2V$



Solⁿ:

$$V_{t,n} = 0.5V, \quad V_{t,p} = 0.2V$$



$$\textcircled{1} V_{sg} = 4 - 3 = 1 > |V_{t,p}| \therefore \text{ON}$$

$$\textcircled{2} V_g = 3V \rightarrow \text{ON}$$

$$V_1: \text{w.r.to } \textcircled{1} V_1 = V_{D1} = 4V$$

$$\text{w.r.to } \textcircled{2} V_1 = V_{S2} = V_{g2} - V_{t,n} = 3 - 0.5 = 2.5V$$

V_1 charges beyond 2.5V through the PMOS, and NMOS becomes OFF. Hence final voltage 4V. $\therefore V_1 = 4V$

$$\textcircled{3} V_g = V_1 = 4V \rightarrow \text{ON}$$

$$V_2 = V_{S3} = V_{g3} - V_{t,n} = 4 - 0.5 = 3.5V \quad \therefore V_2 = 3.5V$$

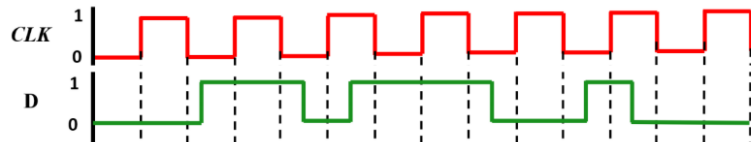
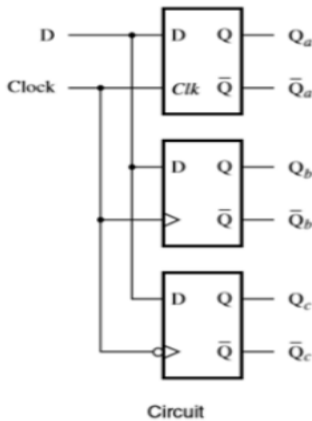
$$\textcircled{4} V_g = 3V \rightarrow \text{ON}$$

$$V_3 = V_{S4} = V_{g4} - V_{t,n} = 3 - 0.5 = 2.5V \quad \therefore V_3 = 2.5V$$

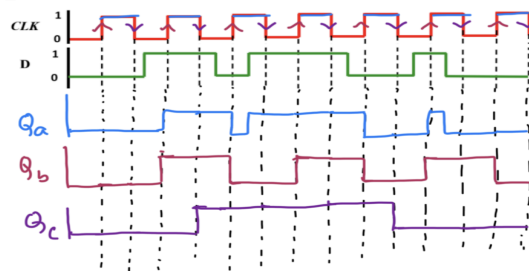
$$\textcircled{5} V_{sg} = 2.5 - 2.5 = 0 < |V_{t,p}| \therefore \text{OFF}$$

$$\therefore V_4 = Z, \quad V_5 = Z$$

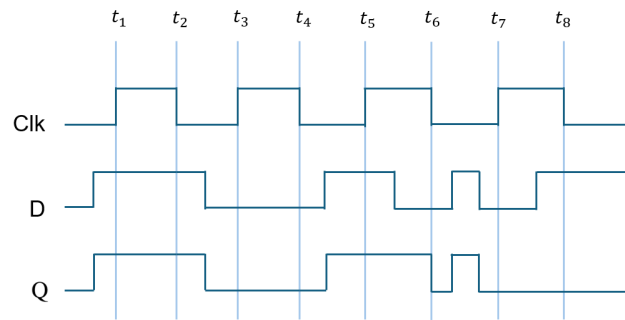
Q-12: For the following circuit, draw Q_a , Q_b and Q_c :



Solⁿ:

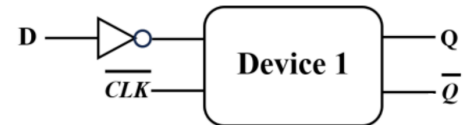
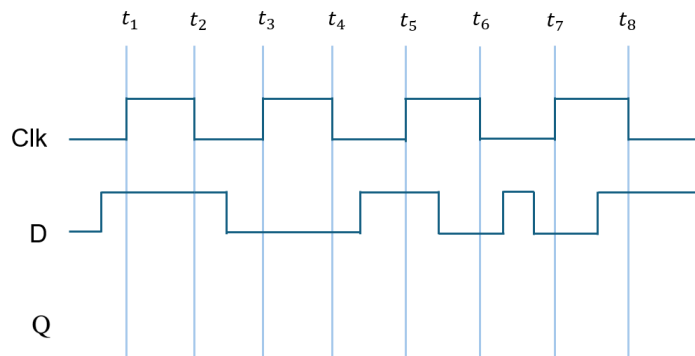


Q-13.



(a) Identify Device-1 and implement it using transmission gates.

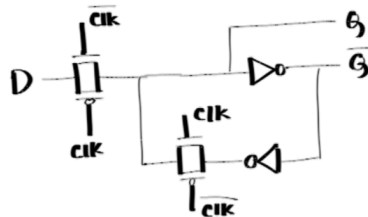
(b) Draw the output waveform based on the following circuit:



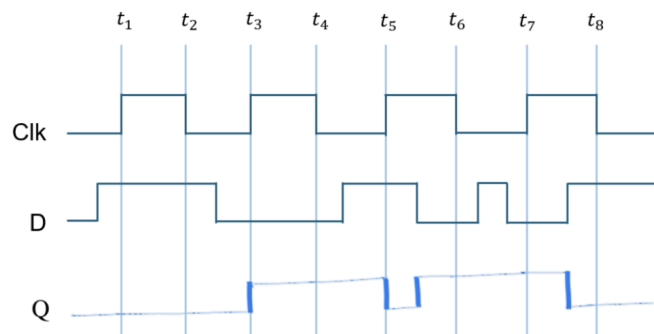
Solution:

a)

-ve D Latch.



b)



Q-14: Identify Device 1 and **draw** the waveforms in the grid given above for the right circuit of Fig. 1. **Determine** if the circuit is a negative-edge/level-triggered or positive-edge/level-triggered flip-flop/latch?

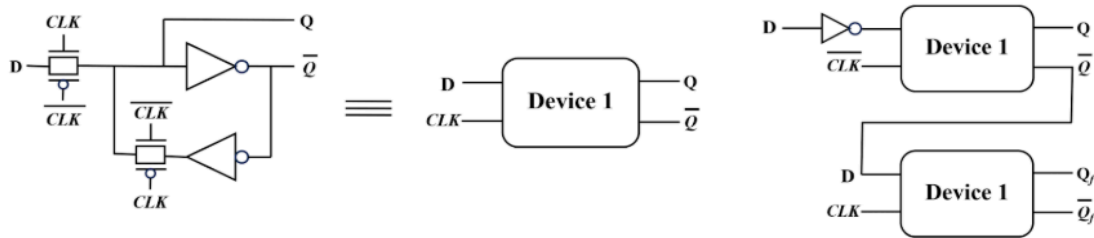
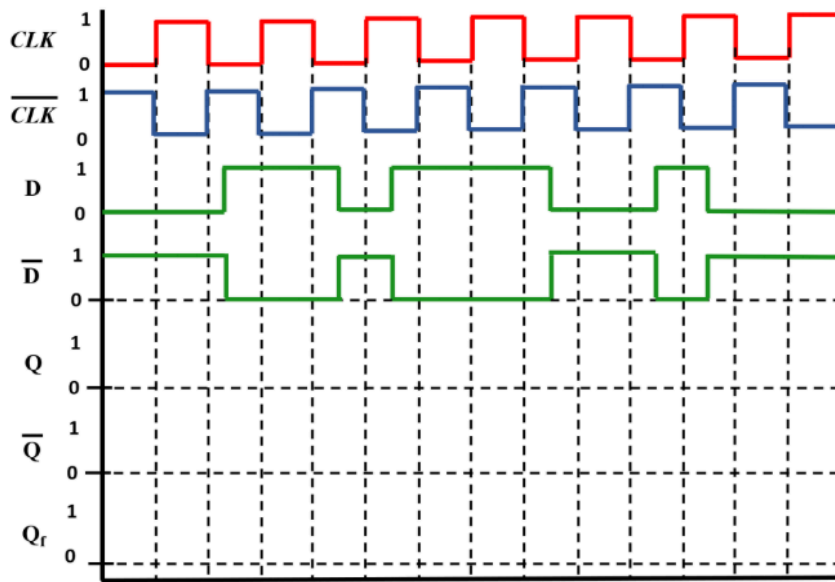
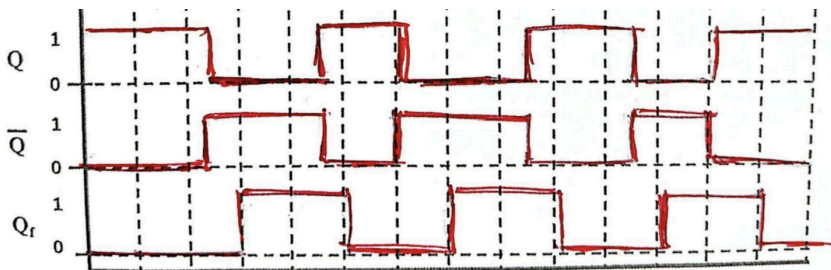


Figure 1: Circuit on the left is represented by the block **Device 1**. Circuit on the right uses this **Device 1**.



Solution:



Q-15.

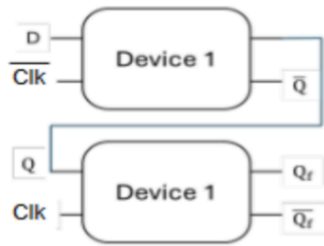


Figure: 1

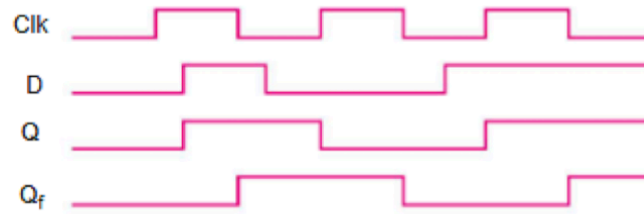


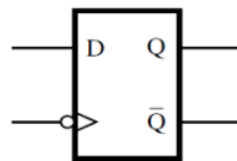
Figure: 2

Figure 2 shows the output waveshape of the device implemented in **Figure 1**.

- A. **Identify** the circuit implemented in **Figure 1** from the timing diagram of **Figure 2**. Also identify **Device 1** shown in **Figure 1**.
- B. **Draw** the circuit of **Device 1** using transmission gates.

Solution: A) Figure-1: Negative edge triggered D Flipflop.
Device-1: Negative level triggered D Latch.

Q-16.



D Flip-flop

Figure 1.1

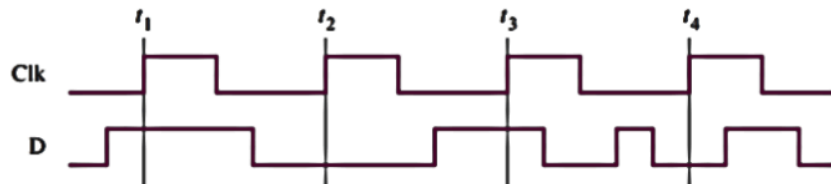


Figure 1.2

(a)	State the difference between D Latch and D Flip-Flop. Implement the D Flip-Flop (Given in the above Figure) using D Latch.
(b)	Draw the output of the D Flip-Flop when Clk and D of Figure 1.2 are applied to the D Flip-Flop.

Solution: Negative edge triggered D Flip Flop

Finite State Machine (FSM)

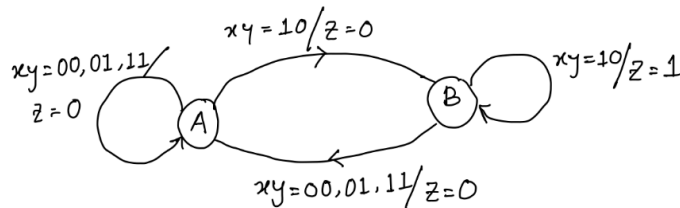
Q-1.

Design a finite state machine (FSM) for a car alarm system. The system has two binary inputs: y , indicating the lock status of the car door ($y = 1$ if **unlocked**, $y = 0$ if **locked**) and x , indicating the door status ($x = 1$ if the **door is open**, $x = 0$ if **closed**). The output z is used to control the alarm: $z = 1$ when the alarm is active, and $z = 0$ otherwise. The alarm should be triggered ($z = 1$) if the door is opened ($x = 1$) while it is still locked ($y = 0$) **for two consecutive clock cycles**. An example is shown below:

y	1	0	1	0	0	0	1	1	0	0	0	0	1	0
x	0	0	1	1	1	0	1	0	1	1	1	0	0	1
z	0	0	0	0	1	0	0	0	0	1	1	0	0	0

(a)	Determine the type of FSM that would be needed for the given problem and design the state diagram for the corresponding FSM.	[2]
(b)	Derive the state-assigned table using the Binary encoding from your state diagram in (a).	[2]
(c)	Find the expression for the next state and the output for the FSM using KMAP .	[3]
(d)	Calculate the number of flip-flops required for the circuit implementation and draw the circuit for this FSM.	[3]

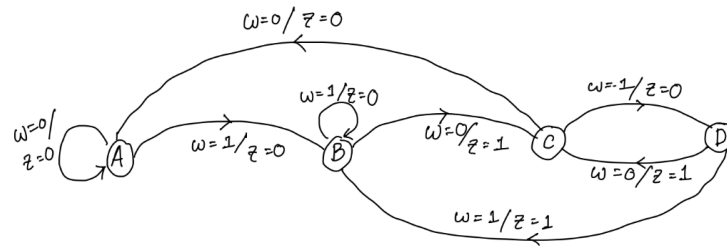
Solⁿ: $xy = 10, 10$ sequence detector. Overlapping Mealy type fsm.



Q-2: Design an FSM (mealy) that detects either '10' or '1011'. Provide

- State diagram
- State Assigned table
- The final circuit using logic gates and FF

Soln: 10 or 1011 sequence detector. Mealy type fsm. Assume over-lapping.



Q-3: A system has one input, A, and two outputs, X and Y.

The output X should be **1** if the input A has been **0** for **three consecutive cycles**.

The output Y should be **1** if the input sequence **010** occurs consecutively. For example,

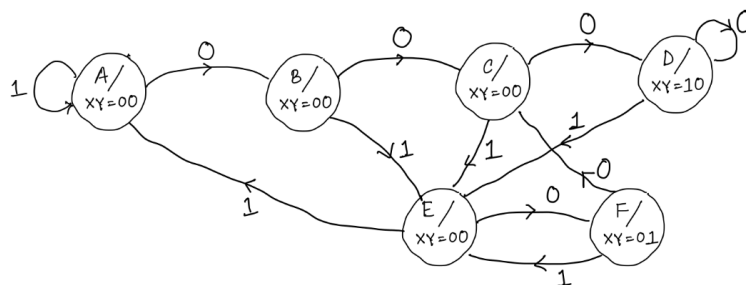
A	1	0	0	0	0	1	0	0	1	0	1	0	0	0
X	0	0	0	0	1	1	0	0	0	0	0	0	0	0
Y	0	0	0	0	0	0	0	1	0	0	1	0	1	0

(a) Draw the state diagram for the corresponding FSM.

(b) Derive the state-assigned table from the state diagram.

(c) Find the expression for the next state and the output for the FSM using KMAP.

Soln: 000 or 010 sequence detector (separate outputs)
Overlapping Moore-type fsm.

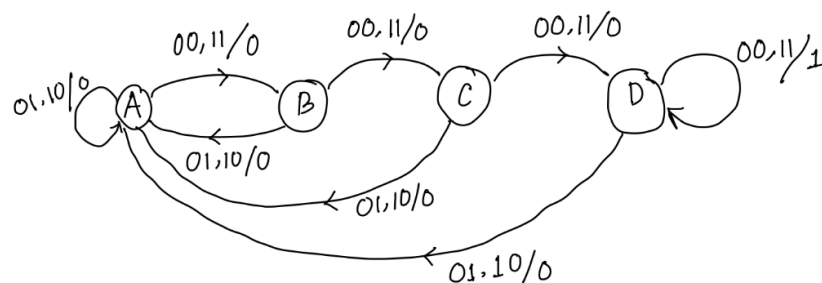


Q-4: A sequential circuit has two inputs, w1 and w2, and an output, z. Its function is to compare the input sequences on the two inputs. If w1 = w2 during any four consecutive clock cycles, the circuit produces z = 1; otherwise, z = 0. For example

w1	0	1	1	0	1	1	1	0	0	0	1	1	0	1
w2	1	1	1	0	1	0	1	0	0	0	1	1	1	0
z	0	0	0	0	1	0	0	0	0	1	1	1	0	0

Draw the **state diagram** of the corresponding FSM.

Solⁿ: 00 or 11 for four consecutive cycle. Overlapping mealy type.



Q-5:

Suppose you want to design an FSM that will detect sequences of alphabet. There will be four different types of possible input alphabets (A, B, C, D). When the sequence of the alphabets is **AACC**, the output of the FSM will be high (1), otherwise it will remain low (0).

Input (w)	A	B	A	A	C	C	A	C
Output (z)	0	0	0	0	0	1	0	0

(a)	Identify the type of FSM and draw the state diagram of the FSM.	[6]
(b)	Make a state assigned table using the gray encoding.	[5]
(c)	Find the expression of the output (z) using the state assigned table from (b).	[4]

soln: AACC sequence detector. Non-overlapping Mealy type fsm.

